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| <b>Name:</b> |  | <b>Index Number:</b> |  | <b>Class:</b> |  |
|--------------|--|----------------------|--|---------------|--|



# **DUNMAN HIGH SCHOOL** **Preliminary Examination** **Year 6**

## **H2 BIOLOGY**

9648/02

Paper 2 Structured and Free-Response Questions

**16 September 2013**

**2 hours**

Additional Materials: Writing paper

### **INSTRUCTIONS TO CANDIDATES:**

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

#### **Section B Structured Questions**

Answer **all** questions.

Write your answers on space provided in the Question Paper.

#### **Section C Free-Response Questions**

Answer **one** question. Your answer to Section C must be in continuous prose, where appropriate. Write your answers on the writing paper provided. Answer (a) and (b) on separate pieces of paper.

**Submit your answers to Sections B and Section C (a) and (b) separately.**

#### **INFORMATION FOR CANDIDATES**

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [ ].

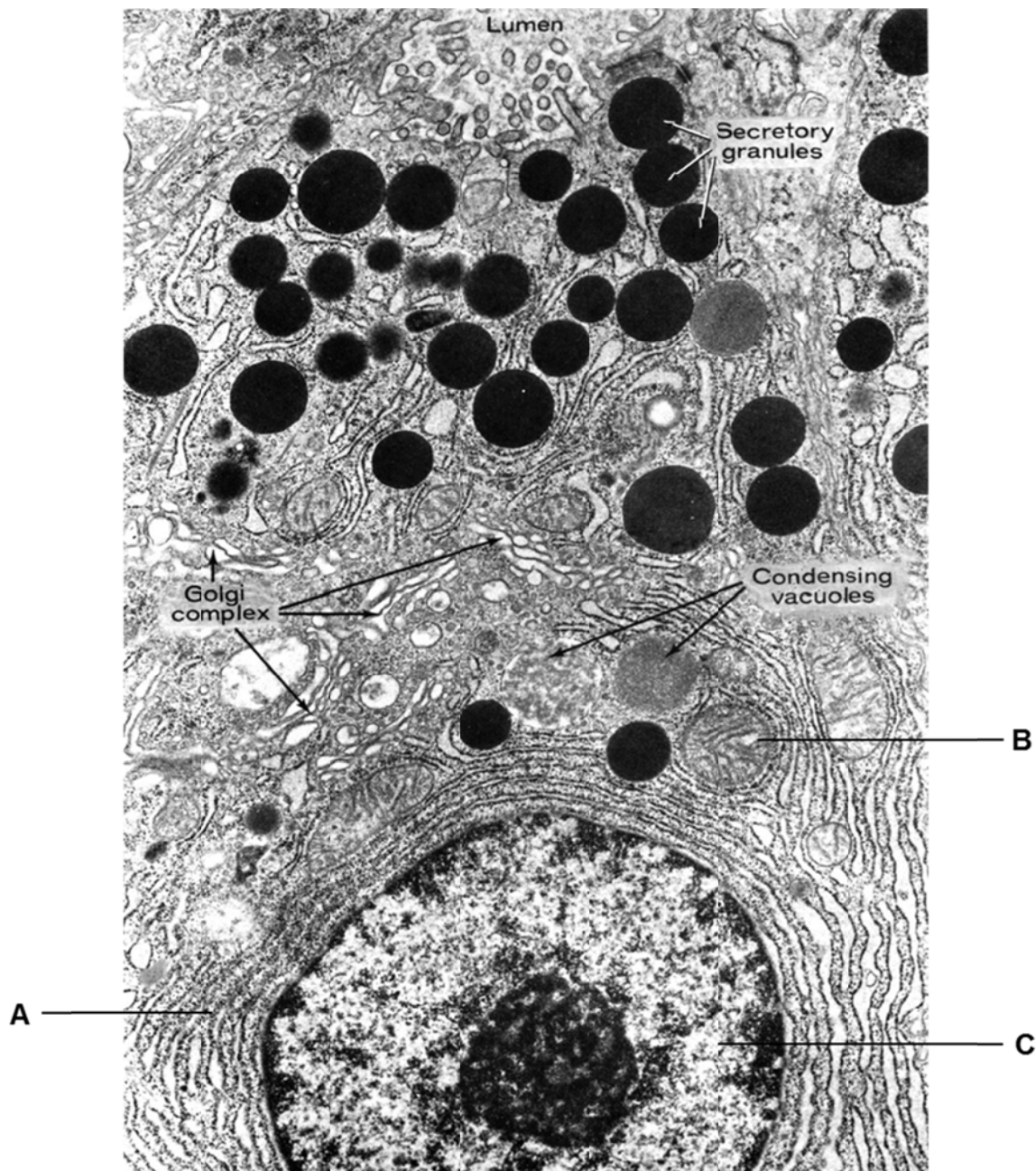
| For Examiner's Use    |             |
|-----------------------|-------------|
| <b>Section A [40]</b> |             |
| <b>Section B [80]</b> |             |
| <b>1</b>              | <b>/ 10</b> |
| <b>2</b>              | <b>/ 5</b>  |
| <b>3</b>              | <b>/ 5</b>  |
| <b>4</b>              | <b>/ 10</b> |
| <b>5</b>              | <b>/ 10</b> |
| <b>6</b>              | <b>/ 9</b>  |
| <b>7</b>              | <b>/ 13</b> |
| <b>8</b>              | <b>/ 9</b>  |
| <b>9</b>              | <b>/ 9</b>  |
| <b>Section C [20]</b> |             |
| <b>1 / 2</b>          |             |
| <b>Total [140]</b>    |             |

This document consists of **23** printed pages and **1** blank page.

**[Turn over**

**Section B: Structured Questions (80 marks)**Answer **all** questions in this section.*For  
Examiner's  
use***Question 1**

**Fig 1.1** below shows a pancreatic acinar cell, an exocrine cell involved in the synthesis and secretion of pancreatic digestive enzymes. It shows the condensing vacuoles in the Golgi region and mature secretory vesicles at the cell apex.

**Fig 1.1**

- (a) Name the organelles labelled **A**, **B** and **C** and explain their role in pancreatic enzyme synthesis. [6]

**A** \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**B** \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

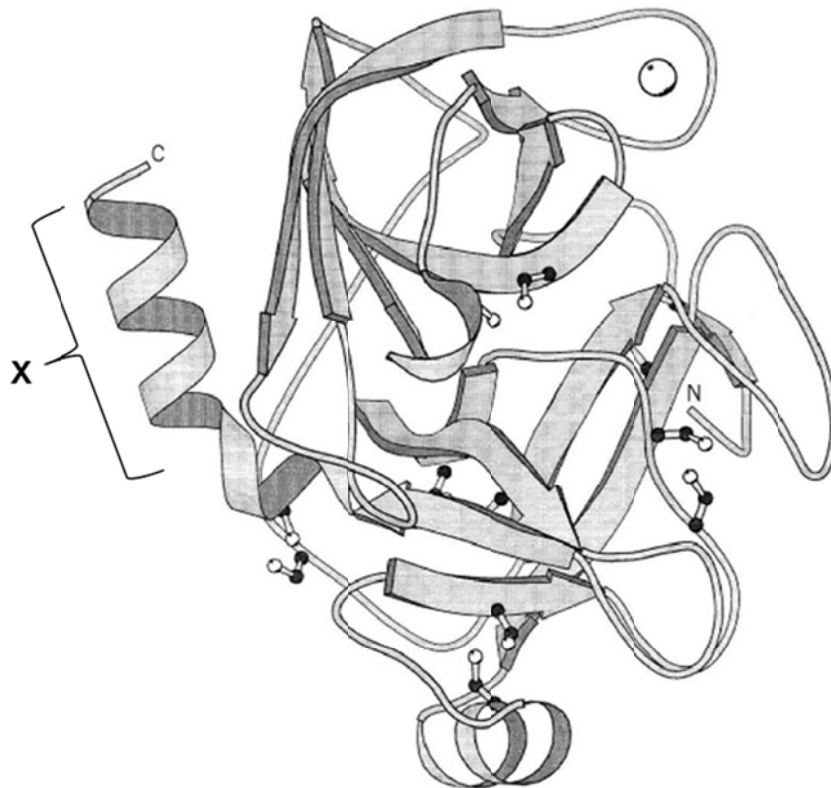
**C** \_\_\_\_\_

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\_\_\_\_\_

- (b) One major pancreatic protease synthesized by the pancreatic acinar cells is **trypsin**, which is synthesized and packaged into secretory vesicles as the inactive proenzyme **trypsinogen**.

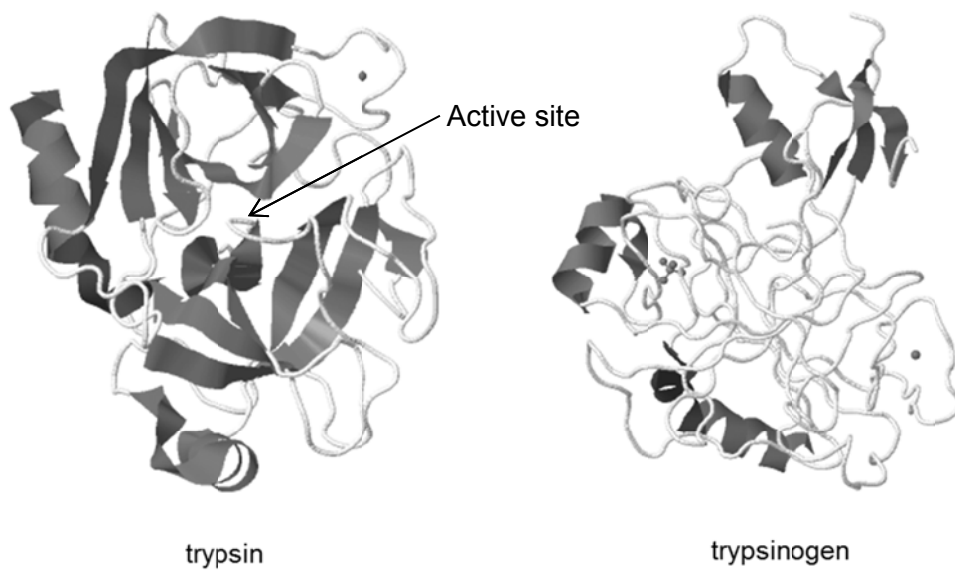
**Fig 1.2** shows the structure of trypsin.



**Fig 1.2**

- (i) Describe region X. [2]

Trypsinogen is converted to trypsin by the removal of a hexapeptide from its amino terminus after it is released into the digestive tract. **Fig 1.3** shows the structure of trypsin and trypsinogen.



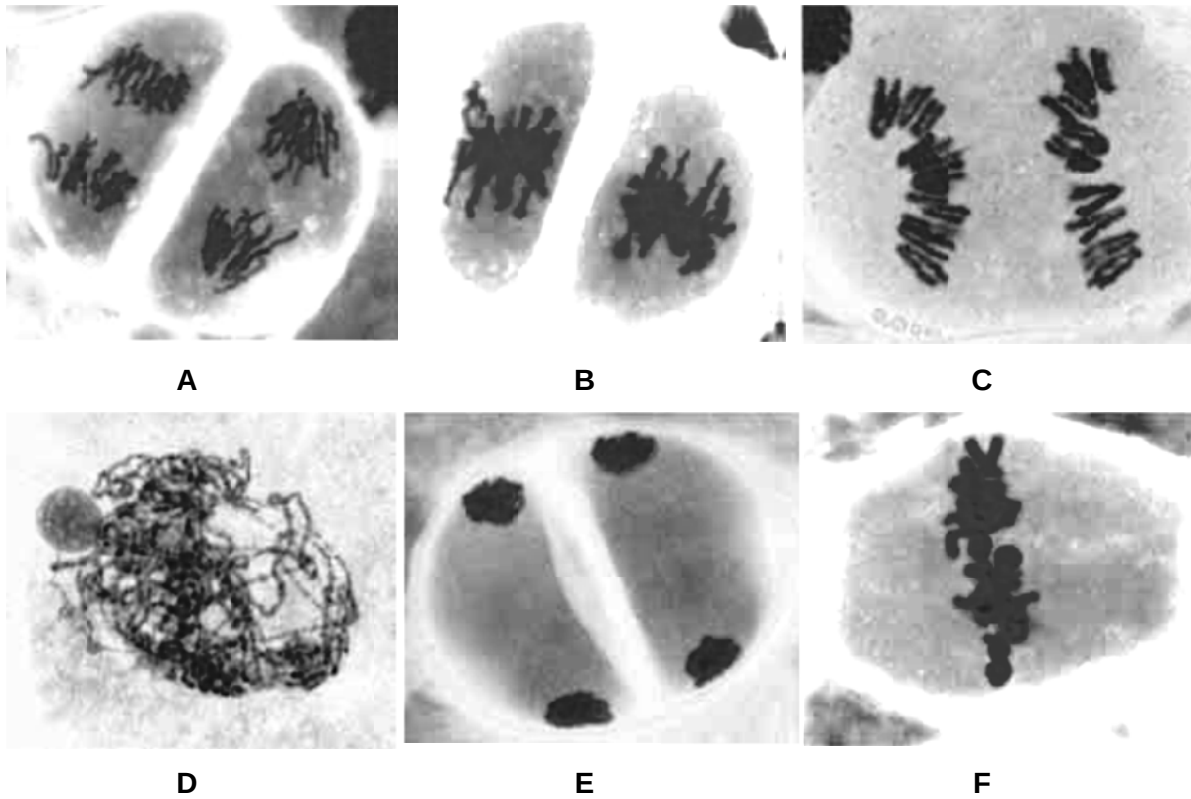
**Fig 1.3**

- (ii) With reference to **Fig 1.3**, explain how the removal of a hexapeptide from the amino terminus of trypsinogen can result in the activation of the enzyme. [2]

**Total: [10]**

**Question 2**

**Fig. 2.1** shows a few stages (A – F) of meiosis in the lily, *Lilium longiflorum*.



**Fig. 2.1**

**(a)** With reference to **Fig. 2.1**,

**(i)** arrange these stages in the correct sequence. [1]

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**(ii)** describe the main events that will occur to complete meiosis after stage A. [2]

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**(b)** Explain how meiosis results in genetically non-identical gametes. [2]

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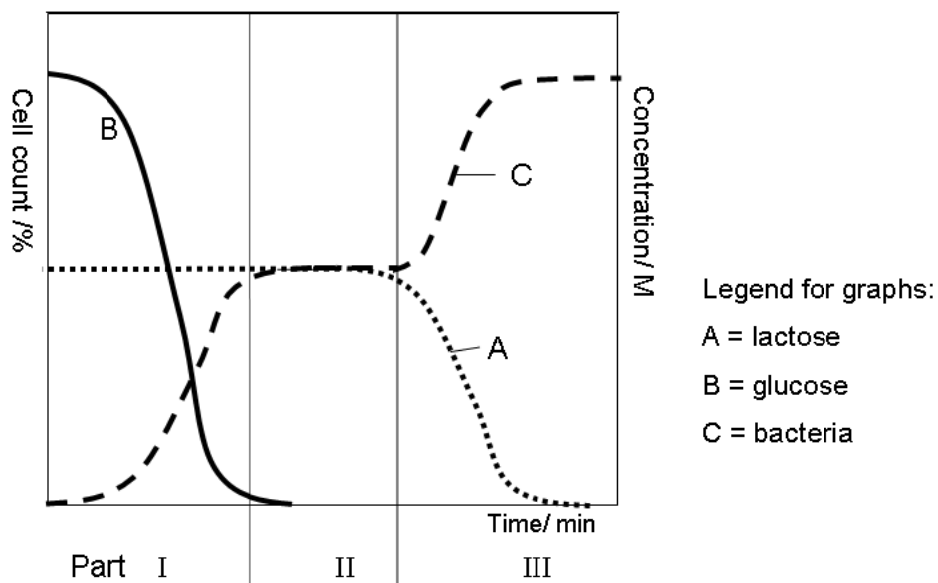
**Total: [5]**

**Question 3**

A study of a bacteria's chromosome showed that it possesses a lac operon to allow the bacterium to metabolise lactose. The structure of the lac operon is shown in **Fig 3.1** below:

**Fig 3.1**

The graph in **Fig 3.2** below shows the growth of bacteria in the presence of glucose and lactose.

**Fig 3.2**

- (a) With reference to **Fig 3.2**, state the main respiratory substrate of the bacteria in **Part I**.  
[1]

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**(b)** Explain the changes in **graph C** in **Part II** and **Part III**. [4]

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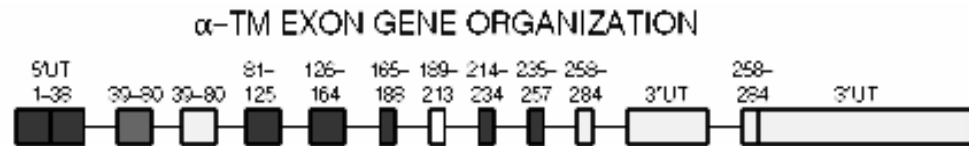
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**Question 4**

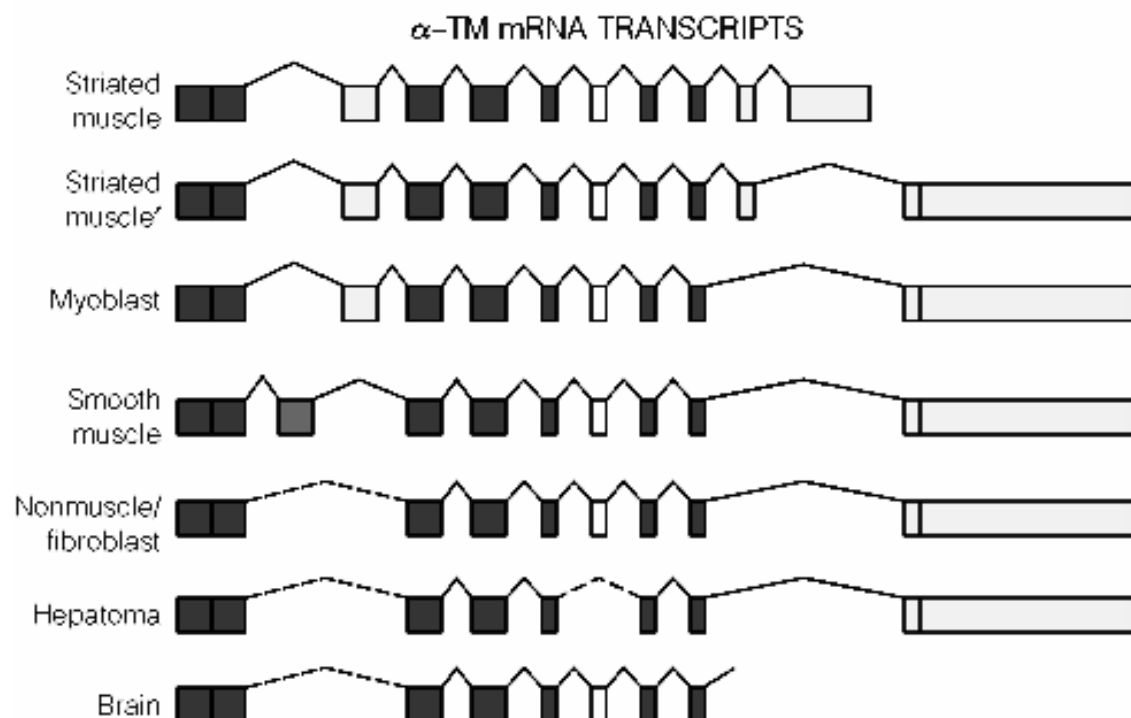
- (a) Alpha tropomyosin ( $\alpha$ -TM) is a protein found in muscles as well as several other tissues in different isoforms.

**Fig 4.1** shows the organisation of the  $\alpha$ -TM gene with exons shown as boxes.



**Fig 4.1**

**Fig 4.2** shows the  $\alpha$ -TM mRNA transcripts found in various tissues.



**Fig 4.2**

- (i) Explain how the  $\alpha$ -TM gene can produce different  $\alpha$ -TM isoforms in different cell types. [3]

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- (ii) Explain the advantage of the process mentioned in (i). [1]

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- (b) Telomeres are found at the ends of eukaryotic chromosomes. They contain a series of tandem repeat sequences and end with a single-stranded section called the 3' overhang. This 3' overhang results from the limitation of DNA polymerase to replicate all the way to the end of the chromosome. This is known as the end-replication problem.

Fig 4.3 shows the structure of a telomere.

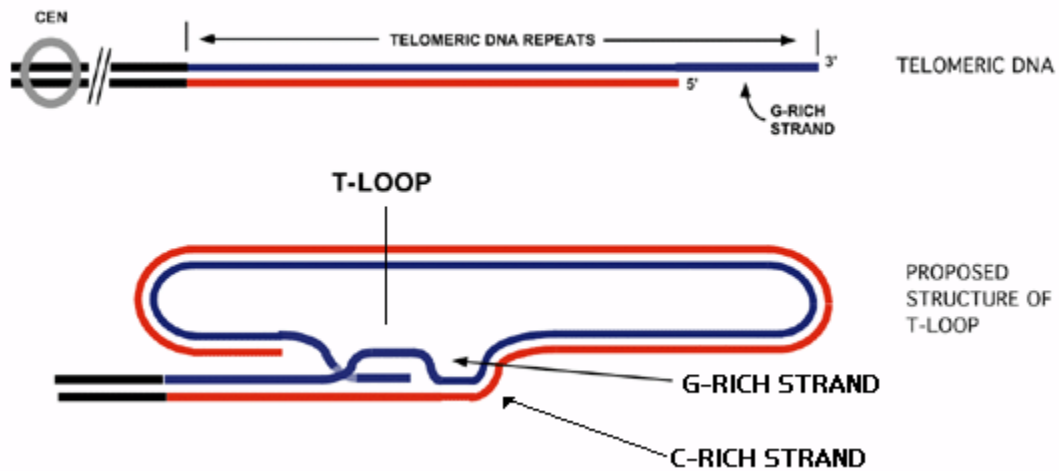


Fig 4.3

- (i) Explain the end-replication problem. [3]

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- (ii) Explain the role of the T loop structure shown in Fig 4.3. [1]

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An enzyme called telomerase catalyses the lengthening of telomeres. Telomerase is typically present only in germ cells and some rapidly dividing somatic cells. It has also been found to be present in high concentrations in cancerous cells.

(iii) Explain how high telomerase activity promotes cancer. [2]

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**Total: [10]**

**Question 5**

In *Drosophila* flies, the wing shape and eye colour are controlled by two different genes found on separate chromosomes. The gene controlling wing shape is found on chromosome 2 and has 2 alleles. Curly wings were found to be dominant over normal wings. **Fig 5.1** and **Fig 5.2** shows the normal wing and curly wing phenotype respectively.

**Fig 5.1: Normal Wing****Fig 5.2: Curly Wing**

The gene controlling eye colour is found on the X chromosome and has two alleles. It was observed that red eye trait is dominant over white eye trait.

- (a) Explain what is meant by the term “**dominant**”? [1]

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- (b) Construct a genetic diagram showing the phenotypes of the offspring and the expected ratio of phenotypes when a white eyed female is crossed with a red eyed male, both are heterozygous for curly wings. [4]

- (c) A geneticist conducted an experiment on the cross mentioned earlier. 240 offspring were obtained from the experiment. The phenotypes of the offspring were as follows:

83 red-eyed females with curly wings  
 41 red-eyed females with normal wings  
 78 white-eyed males with curly wings  
 38 white-eyed males with normal wings

A Chi-square test was performed on the results on the cross.

| Degrees of freedom | Probability, p |      |       |       |       |
|--------------------|----------------|------|-------|-------|-------|
|                    | 0.10           | 0.05 | 0.02  | 0.01  | 0.001 |
| 1                  | 2.71           | 3.84 | 5.41  | 6.64  | 10.83 |
| 2                  | 4.61           | 5.99 | 7.82  | 9.21  | 13.82 |
| 3                  | 6.25           | 7.82 | 9.84  | 11.35 | 16.27 |
| 4                  | 7.78           | 9.49 | 11.67 | 13.28 | 18.47 |

$$\chi^2 \text{ test: } \chi^2 = \sum \frac{(O - E)^2}{E} \quad v = c - 1$$

**Fig 5.3**

- (i) Calculate the Chi-square value, showing your workings in the space below. [2]

- (ii) Use your calculated Chi-square value and the table of probabilities provided in **Fig 5.3** to find the probability that the difference is due to chance. [1]

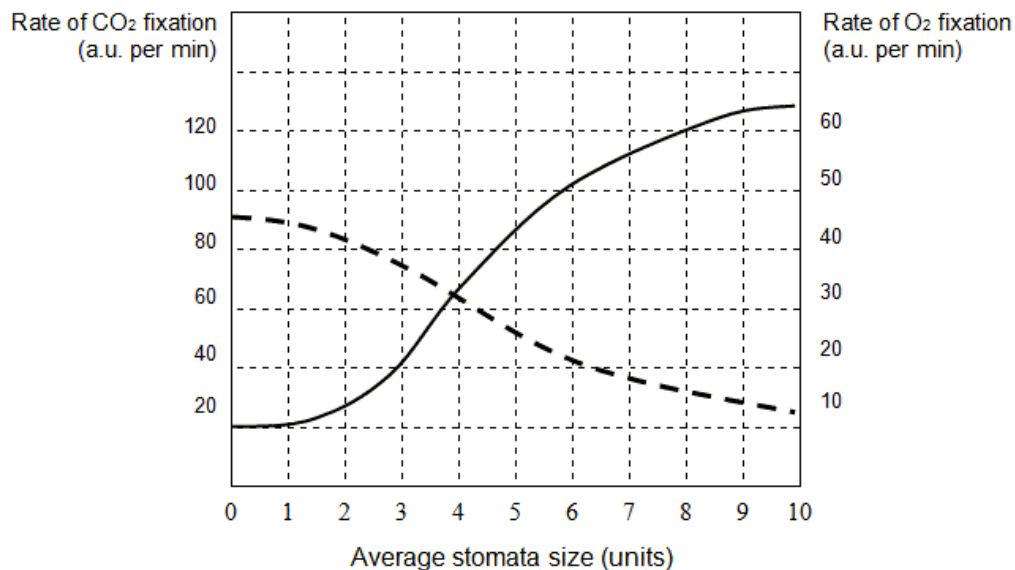
- (iii) State what conclusions may be drawn from the Chi-square test. [2]

**Total: [10]**

**Question 6**

C3 plants undergo photorespiration, a process by which RuBP has oxygen added to it by the enzyme RuBisCO, instead of carbon dioxide during normal photosynthesis. Photorespiration can occur when carbon dioxide levels are low and when temperatures are high, but unlike respiration, photorespiration produces no ATP.

**Fig. 6.1** shows the rate of photosynthesis (measured by rate of CO<sub>2</sub> fixation, graph drawn in solid line) and the rate of photorespiration (measured by rate of O<sub>2</sub> fixation, graph drawn in dotted line) in wheat over a typical summer day.

**Fig. 6.1**

- (a) With reference to **Fig. 6.1**, explain how stomata size affects the overall rate of photosynthesis in wheat. [3]

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- (b) State the type of inhibitor that oxygen is to the enzyme RuBisCO. [1]

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- (c) Some plants have mechanisms to reduce photorespiration, such as the C4 plants. These plants undergo additional steps of capturing carbon dioxide using another enzyme (PEP carboxylase) to form oxaloacetate, before undergoing the Calvin cycle. PEP carboxylase has been found to have a lower  $K_m$  than RuBisCO for carbon dioxide. Examples of C4 plants include sugar cane and maize.

Describe the type of environment that C4 plants are usually found in and explain how PEP carboxylase enables C4 plants to be better adapted to their environments. [3]

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- (d) Explain why Calvin cycle is still necessary in C4 plants. [2]

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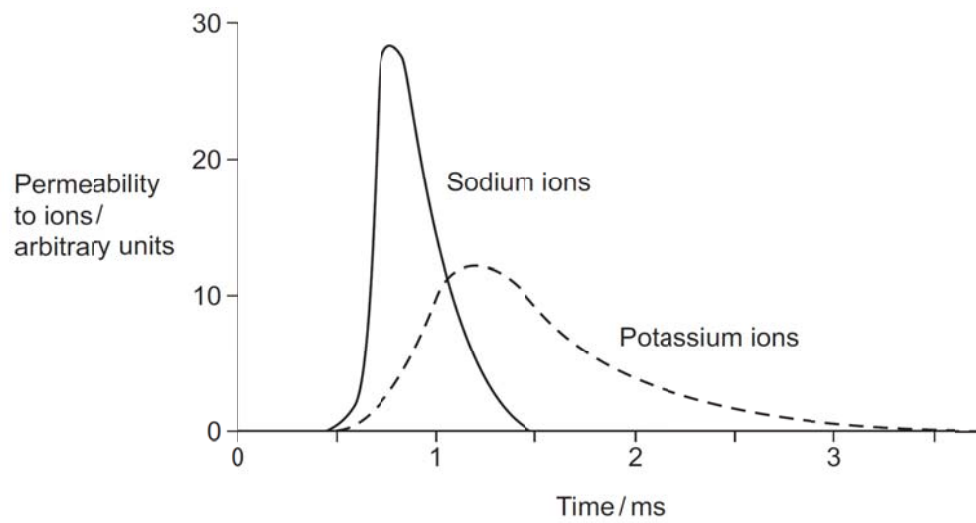
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**Total: [9]**

**Question 7**

**Fig. 7.1** shows the observed changes in permeability of the axonal plasma membrane of a neurone to sodium ions and to potassium ions during a single action potential.



**Fig. 7.1**

- (a) Describe and explain the shape of the curve for sodium ions between 0.5 ms and 0.7 ms. [4]

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- (b) During an action potential, the membrane potential rises to +40 mV and then falls. Use information from **Fig. 7.1** to explain the fall in membrane potential. [3]

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- (c) Tetraethylammonium (TEA) is a drug that blocks voltage-gated potassium channels.

- (i) Predict and explain the effect of TEA on the axonal plasma membrane of the neurone. [2]

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- (ii) Predict and explain the effect of TEA if it could be applied selectively to a presynaptic neurone that releases an excitatory neurotransmitter. [2]

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- (d) After exercise, some ATP is used to re-establish the resting potential in axons. Explain how the resting potential is re-established. [2]

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**Total: [13]**

## Question 8

(a) Fig. 8.1 shows the sequence of events in the inositol phosphate signaling pathway.

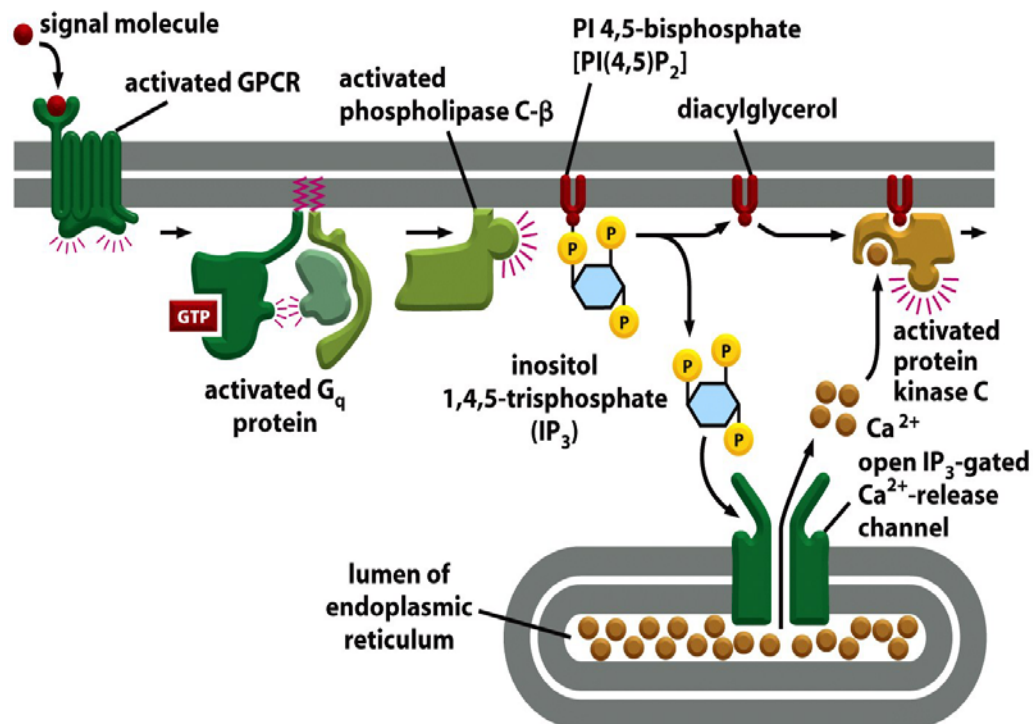


Fig. 8.1

With reference to Fig. 8.1,

- (i) describe the main sequence of events that leads to the activation of protein kinase C. [4]

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- (ii) explain why the signal molecule cannot act directly on the DNA in the nucleus. [1]

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(iii) suggest and explain how an  $\text{IP}_3$ -triggered response is terminated. [2]

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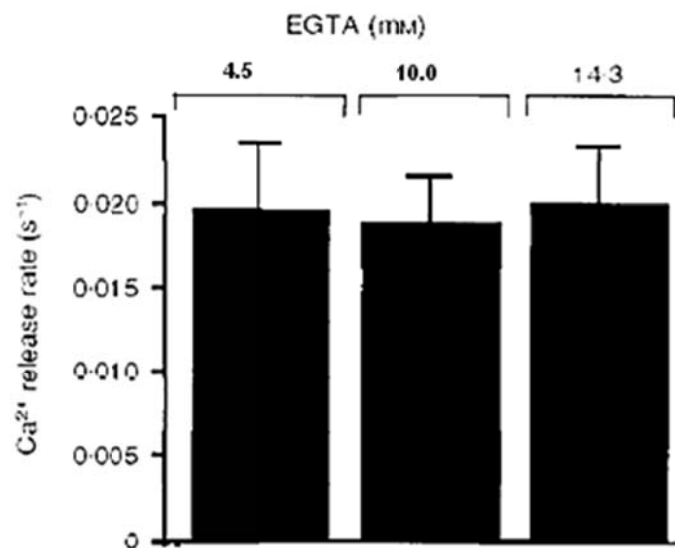
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- (b) Ethylene glycol tetra-acetic acid (EGTA) is a chelating agent that is capable of bonding securely to ions such as  $\text{Ca}^{2+}$ . The pharmacological effects of EGTA have been widely studied.

In one study, smooth muscle cells were transferred to a medium containing varying concentration of EGTA. **Fig. 8.2** shows the effects of EGTA on the released rate of  $\text{Ca}^{2+}$  in an inositol phosphate signalling pathway.



**Fig. 8.2**

Comment if the data in **Fig. 8.2** supports the hypothesis that high concentration of up to 14.3mM of EGTA can effectively inhibit the binding of  $\text{IP}_3$  to its receptor. [2]

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**Total: [9]**

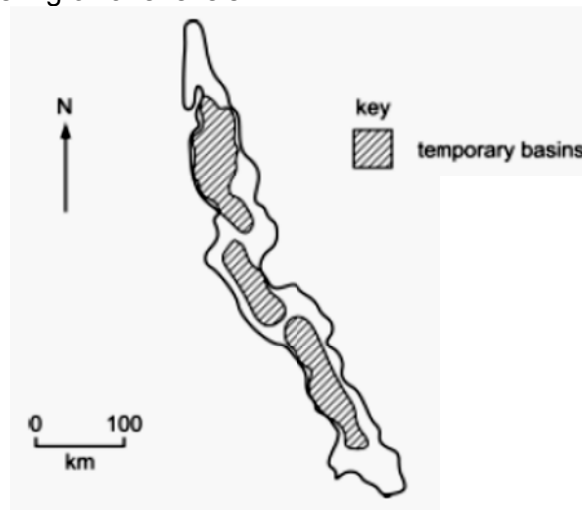
**Question 9**

- (a) In Lake Tanganyika in Africa, there are six species of fish of the genus *Tropheus* and a much larger number of distinctly coloured subspecies of each of the six species. *Tropheus* species are small fish that are confined to isolated rocky habitats around the shores of Lake Tanganyika.

The six species evolved during the primary radiation phase when the lake was first filled, about 1.25 million years ago. They arose from river dwelling ancestors and then filled all available niches in the lake.

Secondary radiations into the many subspecies occurred during the last 200 000 years. Sometime during this period, the water level in the lake fell, resulting in the formation of three separate lake basins. These basins persisted for many thousands of years before the water level rose again.

**Fig. 9.1** shows an outline map of the lake and the location of the three temporary basins caused by lowering of lake levels.



**Fig. 9.1**

Explain how natural selection could have caused the evolution of the six closely related species in the primary radiation. [4]

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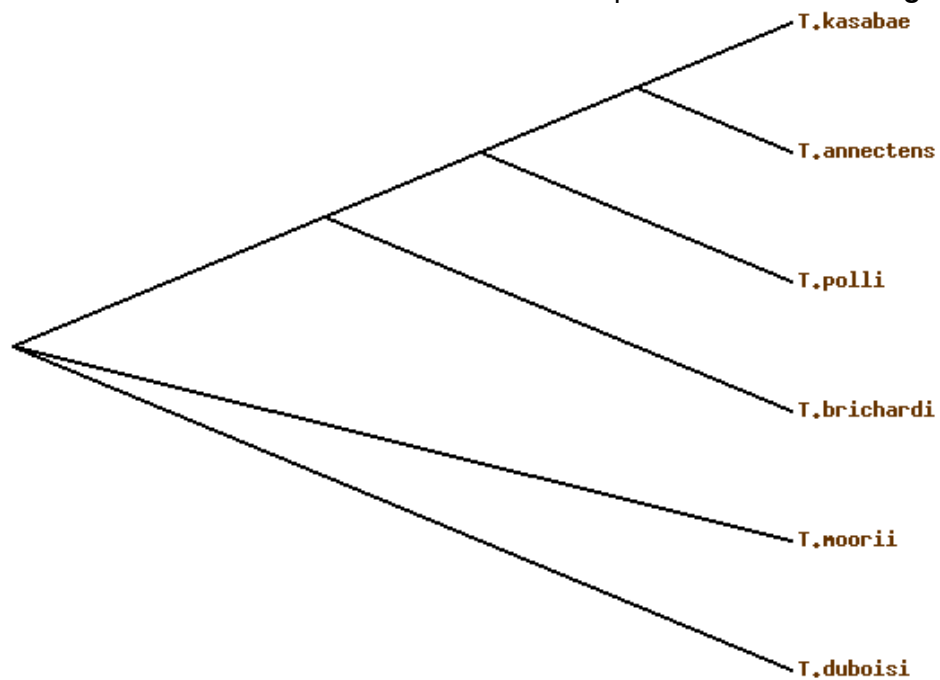
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- (b) Recent research has compared DNA sequence of cytochrome b gene of the six different species of *Tropheus*. **Fig. 9.2** shows part of the cytochrome b DNA sequence.

|                     |                                    |
|---------------------|------------------------------------|
| <i>T. brichardi</i> | CTTTTAACTATGATAACAGCTTTTGTGGGCTACG |
| <i>T. kasabae</i>   | CTTTTAACTATGATAATAGCTTTTGTGGGCTACG |
| <i>T. annectens</i> | CTTTTAACTATGATAACAGCTTTTGTGGGCTACG |
| <i>T. moorii</i>    | CTTTTAACTATGATAACAGCTTTTGTGGGTTACG |
| <i>T. polli</i>     | CTTTTAACTATAATAACAGCTTTTGTGGGCTACG |
| <i>T. duboisi</i>   | CTTTTAACCATGATAACGGCTTTTGTAGGCTATG |

**Fig. 9.2**

A family tree based on differences between the DNA sequences is shown in **Fig. 9.3**.



**Fig. 9.3**

With reference to **Fig. 9.2**, explain why different species of *Tropheus* can have similar DNA sequence in their cytochrome b gene. [2]

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- (c) With reference to **Fig. 9.2** and **Fig 9.3**, describe how the time of divergence of the species could be estimated. [3]

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**Total: [9]**

### Section C: Free-Response Questions (20 marks)

Answer only **one** question.

Write your answers on the writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL RETURN** is required.

#### Question 1

- (a) Compare transcription and translation in eukaryotes. [10]
- (b) (i) Describe how molecular methods can be used to classify organisms. [6]
- (ii) Suggest how bacteria evolve. [4]

**OR**

#### Question 2

- (a) Outline the differences between prokaryotic and eukaryotic control of gene expression. [10]
- (b) (i) Describe Krebs Cycle. [5]
- (ii) Compare oxidative phosphorylation and photophosphorylation. [5]

**Total: [20]**

**Please answer (a) and (b) on separate sheets of writing paper.**

**You will need to hand in (a) and (b) separately.**

**END OF PAPER**

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**DUNMAN HIGH SCHOOL  
PRELIMINARY EXAMINATION 2012  
YEAR SIX  
H2 BIOLOGY (9648)**

## **Answer Scheme**

### **Structured Answers**

#### **Question 1**

**(a)**

**A – Rough Endoplasmic Reticulum;**

Ribosomes on RER responsible for synthesis of proteins (enzymes) to be secreted;  
OR

Packaging of proteins (enzymes) for transport to Golgi Apparatus in transport vesicles;

**B – Mitochondrion;**

Aerobic cellular respiration for the synthesis of ATP necessary for protein synthesis;

**C – Nucleus;**

Contains the genes that code for the various pancreatic enzymes which are transcribed into mRNA for protein synthesis;

**(b)(i)**

Alpha helix,

held by H bonds located between –CO and –NH groups (of the main polypeptide chain);

Between n and n+4 residues in the chain;

**(ii)**

Cleavage of hexapeptide from amino terminus shortens primary structure which changes the secondary structure such that  $\beta$  pleated sheets can form;

this creates a pocket / groove for the substrate to fit into on the surface of the tertiary structure that functions as the active site of the enzyme;

#### **Question 2**

**(a) (i)** D, F, C, B, A, E ;

**(ii)**

- The chromosomes uncoil / decondense and become very indistinct to form chromatin;
- Spindle fibres disassemble / disintegrate / dissassociate;
- Nuclear envelopes reform around each nucleus;

**2 max**

**(b)**

Crossing over between non-sister chromatids / homologous chromosomes during prophase I;

Independent assortment and random segregation of homologous pairs of chromosomes during metaphase I and anaphase I respectively;

Independent assortment and random segregation of non-identical sister chromatids due to crossing over during metaphase II and anaphase II respectively;

**2 max**

**Question 3**

(a) glucose as the level of glucose is decreasing;

(b)

In Part II, there is no increase in the cell count of the bacteria as glucose has been used up and time is needed for the expression/activation of the lac operon to produce  $\beta$ -galactosidase before lactose can be used/metabolized;

The presence of lactose means that the small amount of  $\beta$ -galactosidase present in the bacterial cells will catalyse conversion of some lactose molecules to allolactose;

Allolactose binds to the lac repressor, causing a change in conformation, inactivating it such that it no longer binds to the lac operator. Allowing RNA polymerase access to the promoter;

The absence of glucose means that cAMP levels in the cells are high;

cAMP binds to CAP, causing a change in conformation of CAP such that it binds to the CAP binding site, increasing RNA polymerase's affinity for the promoter, increasing the rate of transcription of  $\beta$ -galactosidase;

many copies of  $\beta$ -galactosidase, allowing bacteria to efficiently use lactose as a respiratory substrate to support growth, thus the increase in the bacterial cell count seen in part III. This continues until lactose is used up and cell count levels off;

**4 max**

**Question 4**

(a)(i)

alternative RNA splicing of the pre-mRNA;

different spliceosomes are present in different cell types which recognize different base sequences are introns and exons;

from 1 gene  $\rightarrow$  1 pre-mRNA  $\rightarrow$  different mRNAs produced  $\rightarrow$  different proteins;

(ii)

1 gene can code for different versions of a protein, this increases the coding capacity of the genome without needing to increase the amount of DNA within the cell;

(b)(i)

DNA polymerase can only add nucleotides to the 3' hydroxyl end of a pre-existing daughter strand / cannot add nucleotides to the bare template;

In lagging strand synthesis, once that primer is removed, it cannot be replaced with DNA as there is no 3' end onto which DNA polymerase can add deoxyribonucleotides;

Therefore the 5' ends of daughter DNA strands of the lagging strand cannot be completed / repeated rounds of replication thus produce shorter and shorter DNA molecules at the 5' end;

(ii)

to prevent exonucleases from degrading the ends of the linear DNA molecules;

to prevent fusions of the ends of chromosome with other DNA molecules;

**1 max**

(iii)

Telomere length plays a role in regulating **replicative cell senescence** as the length of the telomeres limits the number of times each cell can divide / ref to apoptosis;

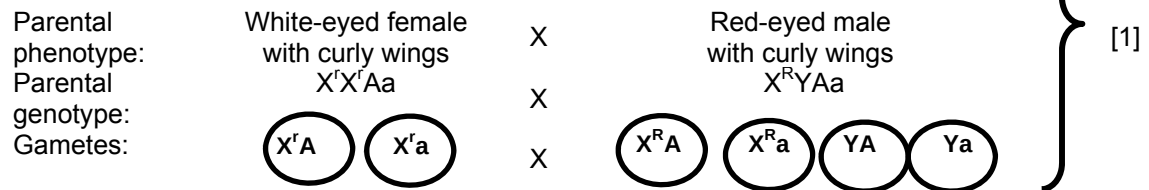
Telomerase allows the generation of telomeres lost at each cell division. This allows cancer cells to divide continuously and indefinitely;

**Question 5****(a)**

In a heterozygous individual, the phenotype of the dominant allele is always expressed and masks the effect of the recessive allele;

**(b)**

Let **A** represent the dominant allele for curly wings  
**a** represent the recessive allele for normal wings  
 $X^R$  represent the dominant allele for red eyes  
 $X^r$  represent the recessive allele for white eyes



Punnett square:

|        | $X^RA$     | $X^Ra$     | $YA$     | $Ya$     |
|--------|------------|------------|----------|----------|
| $X^rA$ | $X^RX^rAA$ | $X^RX^rAa$ | $X^rYAA$ | $X^rYAa$ |
| $X^ra$ | $X^RX^rAa$ | $X^RX^raa$ | $X^rYAa$ | $X^rYaa$ |

|                             |                                              |                                  |                                          |                                  |
|-----------------------------|----------------------------------------------|----------------------------------|------------------------------------------|----------------------------------|
| Offspring genotype:         | $X^RX^rAA$<br><b>2 <math>X^RX^rAa</math></b> | $X^RX^raa$                       | $X^rYAA$<br><b>2 <math>X^rYAa</math></b> | $X^rYaa$                         |
| Offspring phenotype:        | Red eyed female with curly wing              | Red eyed female with normal wing | White eyed male with curly wing          | White eyed male with normal wing |
| Offspring phenotypic ratio: | <b>3 :</b>                                   | <b>1 :</b>                       | <b>3 :</b>                               | <b>1</b>                         |

**(c)(i)**

| Classes                          | Observed number (O) | Expected ratio | Expected number (E) | $(O - E)^2$ | $\frac{(O - E)^2}{E}$ |
|----------------------------------|---------------------|----------------|---------------------|-------------|-----------------------|
| Red eyed female with curly wing  | 83                  | 3              | 90                  | 49          | 0.544                 |
| Red eyed female with normal wing | 41                  | 1              | 30                  | 121         | 4.033                 |
| White eyed male with curly wing  | 78                  | 3              | 90                  | 144         | 1.6                   |
| White eyed male with normal wing | 38                  | 1              | 30                  | 64          | 2.133                 |
| Total                            | 240                 | 8              | 240                 | 0           | 0                     |
|                                  |                     |                |                     | $\Sigma$    | <u>8.31</u>           |

(Calculations / Workings [1] Correct Answer to 2d.p. [1])

(ii)

$0.02 < p < 0.05$ ;

(iii)

Critical  $\chi^2$  value (7.82) < calculated  $\chi^2$  value (8.31), therefore there is a significant difference between observed and expected values;

cross does not follow Mendelian ratios and other factors may be at work;

### Question 6

(a)

As average stomata size increases from 0 to 10 units, rate of photosynthesis increases drastically by 550% / from 20 to 130 a.u.;

An increase in stomata size enables more atmospheric  $\text{CO}_2$  to enter, which is fixed by RuBisCO;

Therefore, overall rate of photosynthesis increases as more  $\text{CO}_2$  is fixed as compared to amount of  $\text{O}_2$  fixed;

(b)

Competitive;

(c)

Dry and hot;

Stomata closes and thus result in lower  $\text{CO}_2$  concentration entering leaves;

Since PEP carboxylase has a higher affinity for carbon dioxide than RuBisCO/ PEP carboxylase not inhibited by oxygen;

Thus C4 plants are able to fix the carbon dioxide more efficiently;

(d)

Calvin cycle is the light-independent stage of photosynthesis;

Only through Calvin cycle can carbon dioxide be fixed to form glucose/ carbohydrate;

**Question 7****(a)**

permeability to sodium ions increases drastically from 0 to 28 a.u;

initial rise in permeability to sodium ions occurs when a stimulus opens some voltage-gated sodium channels;

Influx of sodium ions through those channels depolarises the membrane, making the inside of the neurone positive with respect to the outside / threshold;

As the influx of sodium ions continues, the membrane becomes further depolarised in a positive-feedback loop, opening more voltage-gated sodium channels;

This allows even more sodium ions to diffuse into the neurone, explaining the sharp rise in permeability to sodium ions;

**4 max****(b)**

The fall in membrane potential indicates repolarisation of the membrane as efflux of potassium ions occurs without further influx of sodium ions;

Permeability to sodium ions decreases from 28 to 0 a.u. as most voltage-gated sodium channels are inactivated / Closure of voltage gated sodium channels by the time the membrane potential reaches +40 mV;

Permeability to potassium ions increases up to 12 arbitrary units as most voltage-gated potassium channels are open by the time the membrane potential reaches +40 mV;

**(c)(i)**

Depolarised membrane would not be able to repolarise / return to the resting potential;

This is because efflux of potassium ions cannot occur when voltage-gated potassium channels are blocked by TEA;

**OR**

Takes a long time for resting potential to return back to resting potential;

hence the next action potential will take a longer time to be generated / no action potential can be generated;

**(ii)**

The presynaptic neurone would continue to release excitatory neurotransmitters until the neurotransmitters in the presynaptic neurone are depleted;

This is because calcium ions would continue to enter the presynaptic neurone via the voltage-gated calcium channels and trigger the release of excitatory neurotransmitters so long as the presynaptic membrane remains depolarized;

**OR**

TEA prevents next action potential from being generated since neurone is perpetually in its depolarised state / repolarisation cannot occur when efflux of potassium ions across the voltage gated channels cannot take place.

hence the neurone is not able to release neurotransmitter across the synapse upon stimulation. / action potential cannot be generated / decrease the occurrence of action potential;

**(d)**

Sodium-potassium ion pump by active transport with the hydrolysis of ATP;

2 potassium ions in and 3 sodium ions out of the neurone;

**Question 8****(a) (i)**

Signal molecule binds to GPCR via ligand-receptor interaction which in turn activates G-protein. G-protein is activated when it is attached to GTP nucleotide;

Upon activation, activated G-protein dissociates from the receptor, diffuses along the membrane and then binds to and activates phospholipase C- $\beta$ ;

PI 4,5 biphosphate is cleaved into diacylglycerol and IP<sub>3</sub> / Activation of phospholipase C- $\beta$  catalyzes the formation of IP<sub>3</sub>;

IP<sub>3</sub> serves as the second messenger and binds to ligand-gated Ca<sup>2+</sup> channels in the ER membrane, causing them to open;

Ca<sup>2+</sup> ions flow down concentration gradient from ER lumen to cytosol and binds to protein kinase C, thus activating the enzyme;

**4 max**

**(ii)**

Signal molecule is water soluble / lipid-insoluble, thus it cannot pass through the hydrophobic lipid bilayer/ fatty acid tails of the plasma membrane;

**(iii)**

Dephosphorylation of IP<sub>3</sub> by phosphatase which removes phosphate groups from proteins; IP<sub>3</sub> will no longer be able to bind to ligand-gated Ca<sup>2+</sup> channels, hence leading to the inability of Ca<sup>2+</sup> ions to diffuse into the cytosol;

Presence of calcium ions that enter the cytosol is rapidly pumped out, mainly to the exterior of the cell – disruption to the ionic gradient of Ca<sup>2+</sup> / answers pertaining to the disruption in the Ca<sup>2+</sup> concentration;

Breakdown of signal molecule such that it is unable to bind to GPCR, forming ligand-receptor interaction. Hence, there will be no activation of G-protein and phospholipase C;

**2 max**

**(b)**

Data in Fig. 8.2 do not support the hypothesis that high concentration of EGTA can inhibit the binding of IP<sub>3</sub> to its receptor;

When total EGTA concentration was increased from 4.5 mM to 10.0 mM to 14.3 mM, the calcium release rate remained relatively constant at approximately 0.020 sec<sup>-1</sup>;

**Question 9****(a) (i)**

variations in the Tropheus fish population due to mutation resulting in different alleles;  
 at least 6 different niches in the lake with different selection pressure;  
 fish with at selective advantage survive and reproduce viable offspring;  
 passing on advantageous genes/alleles to the next generation;  
 accumulation of many genetic changes over a long period of time to evolve into 6 species;  
 reproductive/ behavioural/ sympatric/ allopatric/ geographical isolation/ accept hundreds of  
 km apart thus no gene flow between different populations;

**4 max****(b)**

evolved from a common ancestor;  
 cytochrome b gene code for important protein involved in respiration thus any mutation will  
 result in non-functional protein which will be fatal to the organism;  
 this gene would have gone through the least mutation not counting silent mutation;

**3 max****(c)**

compare and note the number of differences in the DNA sequences between species;  
 determine the age of fossil record of each species by dating;  
 calculate the rate of mutation using calibrated molecular clock;  
 no. of mutation/ no. of difference in DNA sequence is used to calculate the length of time  
 since divergence;

**3 max**

## Essay answers

### 1(a)

#### Similarities:

1. Function: Both processes are required for gene expression / protein synthesis;
2. Process: Both involve synthesis of polymers from monomers by condensation reaction with the removal of 1 water molecule;
3. Process: both involve synthesis of polymer based on a reference molecule / template;
4. Process: initiation of both processes requires the aid of factors in formation of initiation complex;
5. Both require post-processing of products – post-transcriptional modification for pre-mRNA to become mature mRNA and post-translational modification of polypeptide chain before formation of mature protein;

#### Differences:

| Features                                 | Transcription                                                                                                     | Translation                                                                                                                     |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 6. Location                              | Nucleus                                                                                                           | Free ribosomes in cytoplasm and ribosomes bonds to the RER                                                                      |
| 7. Template                              | DNA template strand of gene                                                                                       | Mature mRNA                                                                                                                     |
| 8. Direction of reading of template      | Template is read in 3' to 5' direction                                                                            | mRNA is read in 5' to 3' direction                                                                                              |
| 9. Method of reading of template         | Template is read in individual nucleotides, with incoming free ribonucleotide added by complementary base pairing | Template is read in codons, with incoming amino-acyl tRNA anticodon recognizing the codon on mRNA by complementary base pairing |
| 10. Type of monomer                      | Free ribonucleotides                                                                                              | Amino acids                                                                                                                     |
| 11. Monomer requires activation          | no                                                                                                                | Yes<br>Requires activation / attachment to tRNA                                                                                 |
| 12. Bond between basic units             | <u>Phosphodiester bonds</u> between ribonucleotides                                                               | <u>Peptide bonds</u> between amino acids                                                                                        |
| 13. Enzyme (catalyst for bond formation) | RNA polymerase II                                                                                                 | Peptidyl transferase in large ribosomal subunit                                                                                 |
| 14. Products                             | mRNA, rRNA and tRNA                                                                                               | Polypeptide chain                                                                                                               |



**1 (b) (i)**

1. Molecular methods used in classification of organisms involve using molecular homology such as similarities in DNA or amino acid sequences;
2. molecular methods are objective and quantifiable ways to classify organisms base on their phylogenetic relationships;
3. By comparing the DNA or amino acid sequence of different organisms , we can determinate the amount of similarity in the sequences of each birds;
4. the higher the percentage of similarities in DNA or amino acid sequence, the more closely related the organisms are; Accept: vice versa
5. If the species have evolved from the same common ancestors, their DNA or amino acid sequence should show a high degree of homology with each other's and also with that of the original species;
6. based on the idea of molecular clock, different species can be arranged in chronological order of time of evolution, thus creating a phylogeny and extrapolation in phylogenetic studies;
7. Molecular methods can pick up in silent mutation thus the mutation will be reflected in the DNA sequence;
8. New species are evolved when changes in DNA sequence are accumulated over time;

**6 max****(b) (ii)**

Variation in bacteria population due to random gene mutation;

To form new genes;

Lack proof reading mechanism during DNA replication;

Novel genes can also be introduced into bacteria via transformation, transduction and conjugation;

New alleles can be incorporated into the bacteria genome via homologous recombination;

**4 max**

**2(a)****Genetic code**

|                       | <b>Prokaryotes</b> | <b>Eukaryotes</b>                                                                                                                                        |
|-----------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Gene amplification | Does not occur     | Occurs to increase the number of copies of the gene of interest, which hence increases the overall rate of transcription and the amount of gene product. |

**Transcriptional control**

|                                                           | <b>Prokaryotes</b>                                                                                                                                                             | <b>Eukaryotes</b>                                                                                                                                                                                                                            |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2. Access to genes by modification of chromatin structure | Does not occur                                                                                                                                                                 | DNA methylation and histone acetylation / deacetylation can occur, resulting in conversion between euchromatin (transcribed) and heterochromatin (not transcribed).                                                                          |
| 3. Control elements                                       | <ul style="list-style-type: none"> <li>▪ Few</li> <li>▪ Close to promoter / genes under their control</li> <li>▪ E.g. operators located near or within the promoter</li> </ul> | <ul style="list-style-type: none"> <li>▪ More</li> <li>▪ Can be located proximally upstream, or distally upstream / downstream of the gene</li> <li>▪ E.g. enhancer, silencer, proximal control elements, distal control elements</li> </ul> |
| 4. General transcription factors                          | RNA polymerase core enzyme can recognize and bind to the promoter with the aid of the sigma factor                                                                             | General transcription factors required for RNA polymerase to bind to promoter and assembly of transcription initiation complex                                                                                                               |
| 5. Specific transcription factors                         | Repressors bind to operator preventing transcription<br>Activators bind to activator binding sites to increase rate of transcription                                           | <ul style="list-style-type: none"> <li>▪ binding of activators to enhancers to up-regulate transcription</li> <li>▪ repressors binding to silencers / competing with for activator binding sites to down-regulate transcription</li> </ul>   |
| 6. RNA polymerase                                         | All RNAs synthesized by the same RNA polymerase;                                                                                                                               | Three different classes of RNA each synthesized by a different RNA polymerase                                                                                                                                                                |
| 7. Coordinated transcription of genes                     | Genes of same metabolic pathway organized into operons controlled by single promoter, resulting in transcription of single polycistronic mRNA                                  | No operons, each gene has own promoter resulting in monocistronic mRNA,<br>Coordination of gene expression occurs by a combination of control element and specific TFs                                                                       |

**Post-transcriptional**

|                                      | <b>Prokaryotes</b> | <b>Eukaryotes</b>                                                                                             |
|--------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------|
| 8. Post-transcriptional modification | Does not occur     | Addition of 5' 7-methylguanosine cap, 3' poly-A tail, splicing / alternative splicing, editing of mature mRNA |

**Translational**

|                                              | <b>Prokaryotes</b>                                                                                                                                               | <b>Eukaryotes</b>                                                                                                                                                                                                                                     |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9. Simultaneous transcription & translation  | Translation is often begins before transcription is complete as processes are not separated by nuclear envelope                                                  | No direct coupling of transcription and translation. (mRNA must pass across nuclear envelope before translation in the cytoplasm. RNA transcript is not free to associate with ribosomes prior to the completion of transcription).                   |
| 10. Half-life of mRNA                        | Lower stability of transcript / degradation within seconds or minutes / mRNAs shorter half life to rapidly respond to environmental changes                      | Higher stability of transcript, degradation after minutes / days<br><br>controlled by length of poly-A tail (longer tail, slower degradation)                                                                                                         |
| 11. Binding of complementary sequence to RNA | <ul style="list-style-type: none"> <li>Production of anti-sense RNA from non-coding DNA strand can bind to complementary mRNA and inhibit translation</li> </ul> | <ul style="list-style-type: none"> <li>Production of small RNAs (miRNAs and siRNAs) which are cut into smaller pieces by Dicer and picked up by RISC which will degrade any mature mRNA that has sequences complementary to the small RNA.</li> </ul> |
| 12. Ribosome binding sites                   | mRNAs have multiple start codons<br>→ direct the synthesis of several different polypeptides                                                                     | mRNAs have only one start codons → direct synthesis of only one kind of polypeptide                                                                                                                                                                   |

**Post-translational**

|                                     | <b>Prokaryotes</b>                                | <b>Eukaryotes</b>                                                                                                                                           |
|-------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13. Post-translational modification | no/minimal post-translational modifications occur | Post-translational modifications usually required to get functional protein<br><br>e.g. proteolysis, chemical modification, ubiquitination for degradation, |

**2 (b)(i)**

1. Acetyl CoA and oxaloacetate undergo a condensation reaction. Acetyl CoA adds its two-carbon acetyl group to oxaloacetate to form **citrate**. Enzyme = citrate synthetase;
2. Citrate is isomerised to **isocitrate**. Enzyme = aconitase;
3. Isocitrate undergoes oxidative decarboxylation to form  **$\alpha$ -ketoglutarate**. **Carbon dioxide** is lost in this process.  $\text{NAD}^+$  is the hydrogen acceptor and **NADH** is formed. Enzyme = isocitrate dehydrogenase
4.  $\alpha$ -ketoglutarate undergoes oxidative decarboxylation to form **succinyl CoA**. **Carbon dioxide** is lost in this process.  $\text{NAD}^+$  is the hydrogen acceptor and **NADH** is formed. Enzyme =  $\alpha$ -ketoglutarate dehydrogenase
5. Succinyl CoA is converted to **succinate**. **ATP** is formed by substrate level phosphorylation. Enzyme = succinyl CoA synthetase
6. Succinate undergoes dehydrogenation to form **fumarate**. FAD is the hydrogen acceptor and  **$\text{FADH}_2$**  is formed. Enzyme = succinate dehydrogenase
7. Fumarate undergoes hydration to form **malate**. Enzyme = fumarase
8. Malate undergoes dehydrogenation to form **oxaloacetate**.  $\text{NAD}^+$  is the hydrogen acceptor and **NADH** is formed. Enzyme = malate dehydrogenase

**2(b) (ii)****Similarities**

Both processes synthesizes ATP by chemiosmosis, where energy released by the transport of electrons down electron transport chain is used to pump  $\text{H}^+$  against a concentration gradient. And as the  $\text{H}^+$  diffuses back down a concentration gradient, the energy released is used to phosphorylate ADP to ATP;

Both processes involves the transport of electron down ETC.

**2 max****Differences**

| Features                | Oxidative phosphorylation                                                                                                                                                                 | Photophosphorylation                                                                                                                                                                            |   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Definition</b>       | Synthesis of ATP occurs as electrons are transferred from NADH or $\text{FADH}_2$ to oxygen by an <b>electron transport chain</b> . The energy coupling occurs through a proton gradient. | Synthesis of ATP occurs as electrons are transferred from water and chlorophyll molecules to NADP by an <b>electron transport chain</b> . The energy coupling occurs through a proton gradient. | ; |
| <b>Location</b>         | On the inner membrane of mitochondrion.                                                                                                                                                   | on the thylakoid membrane                                                                                                                                                                       | ; |
| <b>Reactions</b>        | Involves a series of redox reactions via a series of electron carriers in the electron transport chain.                                                                                   | hydrolysis of water molecule on top of redox reaction in the electron carriers.                                                                                                                 | ; |
| <b>Synthesis of ATP</b> | by Oxidative phosphorylation                                                                                                                                                              | by both cyclic and noncyclic Photophosphorylation                                                                                                                                               | ; |

**3 max**